Day (37).

(2) Looping or iterative or repetative statements:-

→ The purpose of looping or iterative or repetative statements "to perform certain operation repeatedly for finite number of times until condition Become False."

→ In python programming, we have 2 types of looping statements. They are

(i) ~~[illegible]~~ while loop OR while....else loop

(ii) for loop OR for... else loop.

→ At the time of dealing with looping statements, programmer must ensure 3 points. They are:

(1) Initialization part (from where to start)

(2) Conditional part (where to stop)

(3) Updation part (increment or Decrement)

(1) while Loop OR while.... else loop:-

Syntax:-

```
while (Test Cond):
|--->statement-1
|--->statement-2
|    . . . . . .
|--->statement-n
other statements in program.
```

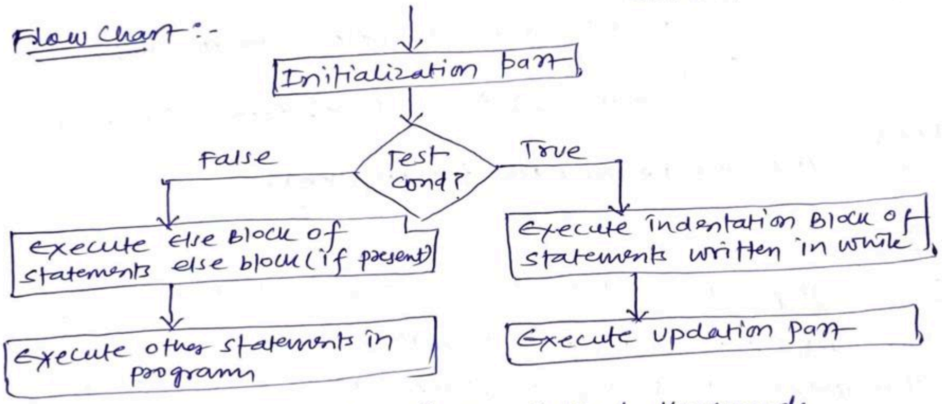
Indentation Block of stmts.

OR

syntax:

```
while (Test Cond):
|--->statement-1
|--->statement-2
|  . . . . .
|--->statement-n
else:
|---->else Block of statement
other statement in prog
```

Flow Chart:-



→ Here 'while' and 'else' are called keywords.

→ Here Test cond can be either True or false.

→ if the Test Cond is True then pvm execute indentation Block of statements and once again pvm control goes to Test cond. if once again Test cond is True then pvm executes indentation Block of statements. This process repeated for finite number of times until Test cond. becomes False.

→ once Test cond becomes False, pvm executes else block of statements which are written else block (if present) and later executes other statements in program.

```
# program for generating 1 to n numbers where
  n is +ve.
n = int(input("Enter How many numbers U want to generate:"))
if (n<=0):
        print("{} is invalid input".format(n))
else:
    i=1         # initialization
    while(i<=n):    # Test cond.
            print("\t{}".format(i))
            i=i+1     # updation
    print("program execution completed")

# program for generating n to 1 numbers where n is +ve.
n = int(input("Enter How many numbers U want to generate:"))
if (n<=0):
      print("{} is invalid input".format(n))
else:
    i=n     # initialization.
    while(i>0):
          print("\t{}".format(i))
          i=i-1
    else:
          print("I am from else part of while loop")
    print("program execution completed")
```

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```
# program for generating mul Table for a given +ve number.
n=int(input("Enter a number for generating Mul table:"))
if(n<=0):
        print("{} is invalid input".format(n))
else:
      print('_'*50)
      print("Mul Table for :{}".format(n))
      print('_'*50)
      i=1        # initialization
      while(i<11):
            print("\t {}x{} ={}".format(n,i,n*i))
            i=i+1
      else:
          print("*"*50)
```

```
# program for generating mul Table for a given +ve number.
n=int(input("Enter a number of generating Mul table:"))
if(n<=0):
       print("{} is invalid input".format(n))
else:
      print("_"*50)
      print("Mul table for {}".format(n))
      print("_"*50)
      for i in range(1,11):
                print("\t {}*{} ={}".format(n,i,n*i))
      else:
          print("*"*50)
```

```
# program for finding sum of first natural numbers
else:
    print ("-" * 50)
    print ("{} first natural Numbers sum".format(n))
    print ("-" *50)
    s=0                                    Sum: n*(n+1)//2
    i = 1
    while (i<=n):
        print ("\t{}".format(i))
        s= s+i
        i = i+1
    else:
        print ("-" *50)
        print ("sum={}".format (s))
        print ("-" *50)
```

```
# program for finding sum of square of first natural
  numbers:-
n=int(input(" Enter How number sum u want to find")

else:
    print ("-" *50)
    print ("\tNatNums\tsquares\tcubes")
    print ("-" *50)
    s, ss, cs = 0,0,0.
    i=1
    while (i<=n)
        print ("\t{}\t\t{}\t\t{}".format
                                    (i,i**2,i**3))
        s =s+i
        ss = ss+i **2
        cs = cs+i **3
        i = i+1
    else: print ("-" *50)
        print ("\t{}\t\t{}\t\t{}".format (s, ss,cs))
```

program for finding product first natural Numbers
Enter How numbers product u want to find.

```
else:
    print ("-"*50)
    print ("product of first {} Natural Nums:". format(n))
    print ("-"*50).
    p=1
    i=1
    while (i<=n)
        print ("\t{}".format (i)))
        p=p*i
        i= i+1
    else:
        print("-"*50)
        print ("product ={}". format (p))
        print ("-"*50)
```

Line into word

```
line =input ("Enter line of Text?")
print ("  —  —  —  —  —")
print ("Given line {}". format(line))
print ("  —  —  —  —")
words = line.split()        # Here words is an obj of list
print ('Given words'={}'. format (words))
print ("—  —  —  —  — — —")
print ("\twords\t\tlength")
print ("—  —  —  —  —  —")
i=0
while(i<len(words)):
        print("\t{} \t\t{}". format(words[i],len(words[i])))
        i=i+1
else:
    print ("  —  —  —  ")
```

1.while Loop or while....else Loop:-

- syntax 1:-
- while (Testcond):
  - stmnt 1
  - stmnt 2
  - ...
  - stmnt n
- other statement in program

syntax 2:-

- syntax 2:-
- while(Testcond):

- - stmnt1
  - stmnt2
  - ....
  - stmntn
- else:
- - else block of stmnt
- other stmnt in program

program for generating 1 to n numbers where n is +ve

In [1]:
```
n=int(input("Enter how many numbers u want to generate :"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    i=1
    while (i<=n):
        print("\t{}".format(i))
        i=i+1
    print("program execution completed")
```

```
	1
	2
	3
	4
	5
	6
	7
	8
program execution completed
```

program for generating n to 1 number when number is positive

In [2]:
```
n=int(input("enter how many number u want generate :"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    i=n
    while (i>0):
        print("\t{}".format(i))
        i=i-1
    else:
        print("i am from else part of the loop")
    print("program execution completed")
```

```
0 is invalid input
```

In [3]:
```
n=int(input("enter how many number u want generate :"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    i=n
    while (i>0):
        print("\t{}".format(i))
        i=i-1
    else:
        print("i am from else part of the loop")
    print("program execution completed")
```

```
-9 is invalid input
```

In [4]:

```
n=int(input("enter how many number u want generate :"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    i=n
    while (i>0):
        print("\t{}".format(i))
        i=i-1
    else:
        print("i am from else part of the loop")
    print("program execution completed")
```

```
        8
        7
        6
        5
        4
        3
        2
        1
i am from else part of the loop
program execution completed
```

program for generating mul table for a given +ve number

In [10]:

```
n=int(input("enter a number for generating mul table:"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    print("="*50)
    print("MUL TABLE for:{}".format(n))
    print("="*50)
    i=1
    while(i<11):
        print("\t {}*{}={}".format(n,i,n*i))
        i=i+1
    else:
        print("="*50)
```

```
==================================================
MUL TABLE for:11
==================================================
         11*1=11
         11*2=22
         11*3=33
         11*4=44
         11*5=55
         11*6=66
         11*7=77
         11*8=88
         11*9=99
         11*10=110
==================================================
```

In [15]:

```
#2nd method
n=int(input("enter a given number of generating mul table:"))
if (n<=0):
    print("{} is invalid input".format(n))
```

```
else:
    print("=="*50)
    print("mul table for {}".format(n))
    print("=="*50)
    for i in range(1,11):
        print("\t{}*{}={}".format(n,i,n*i))
    else:
        print("=="*50)
```

```
====================================================================================================
mul table for 11
====================================================================================================
        11*1=11
        11*2=22
        11*3=33
        11*4=44
        11*5=55
        11*6=66
        11*7=77
        11*8=88
        11*9=99
        11*10=110
====================================================================================================
```

program for findng sum of first natural number:-

In [2]:

```
n=int(input("enter how number sum u want to find :"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    print("=="*50)
    print("\t Natural numbers")
    print("=="*50)
    s=0
    i=1
    while(i<=n):
        print("\t{}".format(i))
        s=s+i
        i=i+1
    else:
        print("=="*50)
        print("sum={}".format(s))
        print("=="*50)
```

```
====================================================================================
===================
         Natural numbers
====================================================================================
===================
        1
        2
        3
        4
        5
====================================================================================
===================
sum=15
====================================================================================
===================
```

program for finding sum of squares of first natural nummbers:

In [9]:
```
n=int(input("enter how number sum u want to find"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    print("=="*50)
    print("\t natnums \t squares \cubes")
    print("=="*50)
    s,ss,cs=0,0,0
    i=1
    while(i<=n):
        print("\t\t{}\t\t{}\t\t{}".format(i,i**2,i**3))
        s=s+i
        ss=ss+i**2
        cs=cs+i**3
        i=i+1
    else:
        print("=="*50)
        print("\t{}\t\t{}\t\t{}".format(s,ss,cs))
```

```
====================================================================================
===================
         natnums         squares \cubes
====================================================================================
===================
                1               1               1
                2               4               8
                3               9               27
                4               16              64
                5               25              125
====================================================================================
===================
        15              55              225
```

program for finding product first natural numbers:-

In [10]:
```
n=int(input("enter how numbers product u want to find"))
if(n<=0):
    print("{} is invalid input".format(n))
else:
    print("=="*50)
    print("product of first {} natural numbers:".format(n))
```

```
    print("=="*50)
    p=1
    i=1
    while(i<=n):
        print("\t{}".format(i))
        p=p*i
        i=i+1
    else:
        print("=="*50)
        print("product={}".format(p))
        print("=="*50)
```

```
=================================================================================
===================
product of first 4 natural numbers:
=================================================================================
===================
        1
        2
        3
        4
=================================================================================
===================
product=24
=================================================================================
===================
```

line into word

In [16]:

```
line=input("enter line of text:")
print("====================================")
print("given line:{}".format(line))
words=line.split()
print("given words={}".format(words))
print("=====================================")
print("\twords\t\tlength")
print("=====================================")
i=0
while(i<len(words)):
    print("\t{}\t\t{}".format(words[i],len(words[i])))
    i=i+1
else:
    print("====================================")
```

```
====================================
given line:python is an oop lang.
given words=['python', 'is', 'an', 'oop', 'lang.']
=====================================
        words           length
=====================================
        python          6
        is              2
        an              2
        oop             3
        lang.           5
====================================
```

In [ ]: